GRAND CANYON NATIONAL PARK
WORLD HERITAGE SITE
REHABILITATE NORTH RIM INFRASTRUCTURE
VOLUME 5 – ROARING SPRINGS PUMPHOUSE (RSPH)

VOLUME #	DESCRIPTION
VOLUME 0	GENERAL
VOLUME 1	PIPELINES - NORTH RIM
VOLUME 2	PIPELINES - NORTH RIM TO ROARING SPRINGS PUMPHOUSE (RSPH)
VOLUME 3	PIPELINES - NORTH RIM TO SUPAI TUNNEL
VOLUME 4	PIPELINES - ROARING SPRINGS CAVE TO ROARING SPRINGS PUMPHOUSE (RSPH)
VOLUME 5	ROARING SPRINGS PUMPHOUSE (RSPH)
VOLUME 6	NORTH RIM WATER TREATMENT PLANT
VOLUME 7	NOT USED
VOLUME 8	NORTH RIM WASTEWATER COLLECTION SYSTEM
VOLUME 9	POWER DISTRIBUTION - NORTH RIM
VOLUME 10	POWER DISTRIBUTION - NORTH RIM TO ROARING SPRINGS PUMPHOUSE (RSPH)
VOLUME 11	POWER DISTRIBUTION - ROARING SPRINGS PUMPHOUSE (RSPH) TO MANZANITA
VOLUME 12	POWER SYSTEM - COTTONWOOD CAMPGROUND

SHEET	SUB SHEET	SUB AREA	TITLE OF SHEET
1	050G001	GENERAL	SHEET INDEX
2	050C101	GENERAL	SITE PLAN
3	050S001	GENERAL	GENERAL STRUCTURAL NOTES
4	050D101	GENERAL	RAW WATER HYDRAULIC PROFILE
5	050D102	GENERAL	FINISHED WATER HYDRAULIC PROFILE
6	050D601	GENERAL	PROCESS FLOW DIAGRAM
7	050D602	GENERAL	DESIGN CRITERIA
8	050D603	GENERAL	PIPE SCHEDULE
9	050Y601	GENERAL	SCADA NETWORK BLOCK DIAGRAM
10	051X101	SITEWORK	DEMOLITION PLAN
11	051C101	SITEWORK	GESC PLAN
12	051C102	SITEWORK	CIVIL SHEET MAP
13	051C103	SITEWORK	OVERALL CIVIL SITEWORK PLAN
14	051C104	SITEWORK	PLAN AND PROFILE A
15	051C105	SITEWORK	PLAN AND PROFILE B+C
16	051C106	SITEWORK	PLAN AND PROFILE D
17	051C107	SITEWORK	PLAN AND PROFILE E
18	051C108	SITEWORK	RSPH UTILITY CORRIDOR ENLARGED PLAN
19	051C301	SITEWORK	RSPH UTILITY SECTIONS
20	051C401	SITEWORK	SLOPE STABILIZATION ENLARGED SITE PLAN
21	051C501	SITEWORK	CIVIL DETAILS TRENCHING
22	051C502	SITEWORK	CIVIL DETAILS PIPE ANCHORS
23	051C503	SITEWORK	CIVIL DETAILS HIGH TENSILE WIRE MESH
24	051C504	SITEWORK	CIVIL DETAILS
25	051C505	SITEWORK	ISOLATION VALVE VAULT
26	051C506	SITEWORK	CIVIL DRAIN FIELD DETAILS
27	051C901	SITEWORK	PHOTO DETAILS
28	051S101	SITEWORK	STUCTURAL / GEOTECHNICAL SITE PLAN
29	051S501	SITEWORK	STRUCTURAL DETAILS
30	051E101	SITEWORK	ELECTRICAL SITE PLAN
31	051E501	SITEWORK	TRANSFORMER AND TRENCH DETAIL
32	051E502	SITEWORK	NOT USED
33	052X101	DEMOLITION RSPH	ARCHITECTURAL DEMOLITION PLAN
34	052X102	DEMOLITION RSPH	ARCHITECTURAL ROOF DEMOLITION PLAN
35	052X103	DEMOLITION RSPH	DEMOLITION PLAN STRUCTURAL
36	052X104	DEMOLITION RSPH	DEMOLITION PLAN PROCESS
37	052X105	DEMOLITION RSPH	DEMOLITION PLAN MECHANICAL
38	052X106	DEMOLITION RSPH	DEMOLITION PLAN ELECTRICAL
39	052X301	DEMOLITION RSPH	DEMOLITION SECTIONS PROCESS
40	052X601	DEMOLITION RSPH	DEMOLITION MCC ONELINE
41	052X602	DEMOLITION RSPH	DEMOLITION MCC ELEVATION
42	052F001	RSPH	CODE ANALYSIS
43	052F101	RSPH	LIFE SAFETY PLAN
44	052S101	RSPH	FLOOR PLAN
45	052S102	RSPH	ROOF FRAMING PLAN
46	052S301	RSPH	SECTIONS
47	052S401	RSPH	ENLARGED FLOOR PLAN
48	052S501	RSPH	DETAILS
49	052A101	RSPH	FLOOR PLAN
50	052A102	RSPH	ROOF PLAN
51	052A201	RSPH	EXTERIOR ELEVATIONS
52	052A202	RSPH	EXTERIOR ELEVATIONS
53	052A301	RSPH	BUILDING SECTIONS
54	052A302	RSPH	WALL SECTION DETAILS
55	052A501	RSPH	INDUSTRIAL CURTAIN DETAIL
56	052A901	RSPH	3D REPRESENTATION
57	052D101	RSPH	PROCESS FLOOR PLAN
58	052D102	RSPH	PROCESS ENLARGED PLANS
59	052D103	RSPH	PROCESS ENLARGED PLANS
60	052D104	RSPH	PROCESS ENLARGED PLAN
61	052D105	RSPH	PROCESS ENLARGED PLAN
62	052D301	RSPH	PROCESS SECTION
63	052D302	RSPH	PROCESS SECTIONS
64	052D303	RSPH	PROCESS SECTION
65	052D304	RSPH	PROCESS SECTION
66	052D305	RSPH	PROCESS SECTIONS
67	052D306	RSPH	PROCESS SECTION
68	052D307	RSPH	PROCESS SECTION
69	052D308	RSPH	PROCESS SECTIONS
70	052D309	RSPH	PROCESS SECTION
71	052D310	RSPH	PROCESS SECTIONS
72	052D501	RSPH	PROCESS DETAILS
73	052D502	RSPH	PROCESS DETAILS
74	052D503	RSPH	TANK DETAILS
75	052D601	RSPH	VALVE DESIGN CRITERIA
76	052D901	RSPH	BLADDER TANK AND BOOSTER PUMPS ISOMETRIC
77	052D902	RSPH	AUTO FILTERS ISOMETRIC
78	052D903	RSPH	RAW WATER PUMPS ISOMETRIC
79	052D904	RSPH	CHEMICAL AREA ISOMETRIC
80	052D905	RSPH	FINISHED WATER STORAGE TANK ISOMETRIC
81	052M101	RSPH	HVAC PLAN
82	052M301	RSPH	HVAC SECTIONS
83	052M601	RSPH	HVAC SCHEDULES
84	052M701	RSPH	HVAC CONTROLS
85	052P101	RSPH	PLUMBING PLAN
86	052P301	RSPH	PLUMBING SECTIONS
87	052P501	RSPH	PLUMBING DETAILS
88	052P601	RSPH	PLUMBING SCHEDULES
89	052E111	RSPH	ELECTRICAL POWER PLAN
90	052E121	RSPH	LIGHTING AND RECEPTACLE PLAN
91	052E131	RSPH	SPECIAL SYSTEMS
92	052E301	RSPH	SERVICE ENTRANCE
93	052E601	RSPH	ONE-LINE DIAGRAM MCC
94	052E602	RSPH	VFD 52211 AND VFD 52212 ONE-LINE DIAGRAMS
95	052E603	RSPH	MISCELLANEOUS ONE-LINE DIAGRAMS
96	052E604	RSPH	MISCELLANEOUS ONE-LINE DIAGRAMS
97	052E605	RSPH	MCC ELEVATION
98	052E606	RSPH	RSPH PLC CP ONE LINE DIAGRAM SHEET 1
99	052E607	RSPH	RSPH PLC CP ONE LINE DIAGRAM SHEET 2
100	052E631	RSPH	ELECTRICAL LUMINAIRE SCHEDULES
101	052E632	RSPH	ELECTRICAL PANELBOARD SCHEDULES
102	052E641	RSPH	MOTOR CONTROL DIAGRAMS
103	052E642	RSPH	MOTOR CONTROL DIAGRAMS
104	052E643	RSPH	MOTOR CONTROL DIAGRAMS
105	052E901	RSPH	ISO CONDUITS FOR PMP-52211 AND PMP-52212
106	052Y501	RSPH	INSTRUMENTATION DETAILS
107	052Y601	RSPH	RAW WATER PUMPING P&ID
108	052Y602	RSPH	FINISHED WATER CONVEYANCE P&ID
109	052Y603	RSPH	CHLORINE DOSING P&ID
110	052Y604	RSPH	FINISHED WATER BOOSTER PUMPS P&ID
111	052Y605	RSPH	MISCELLANEOUS SYSTEMS P&ID
112	052Y612	RSPH	LCP-53117 VALVE CONTROL PANEL LAYOUT
113	052Y613	RSPH	LCP-53117 VALVE CONTROL PANEL POWER SCHEMATIC

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 10-9-2026 Date Signed ARIZONA, U.S.A.

Nicholas Humphrey

REGISTERED PROFESSIONAL ENGINEER (CIVIL) CERTIFICATE NO. 41809 RYAN RHOADES 10-9-26 Date Signed ARIZONA, U.S.A.

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	D. ZINCHENKO	050 G001	GENERAL SHEET INDEX
	CADD N. LYNCH		
	TECH. REVIEW: GOUGH/GLOVER		
	DATE: AUG 2026		NORTH RIM - GRAND CANYON NATIONAL PARK

DRAWING NO.	113 182752
PMIS/PKG NO.	318719
SHEET	1 OF 113

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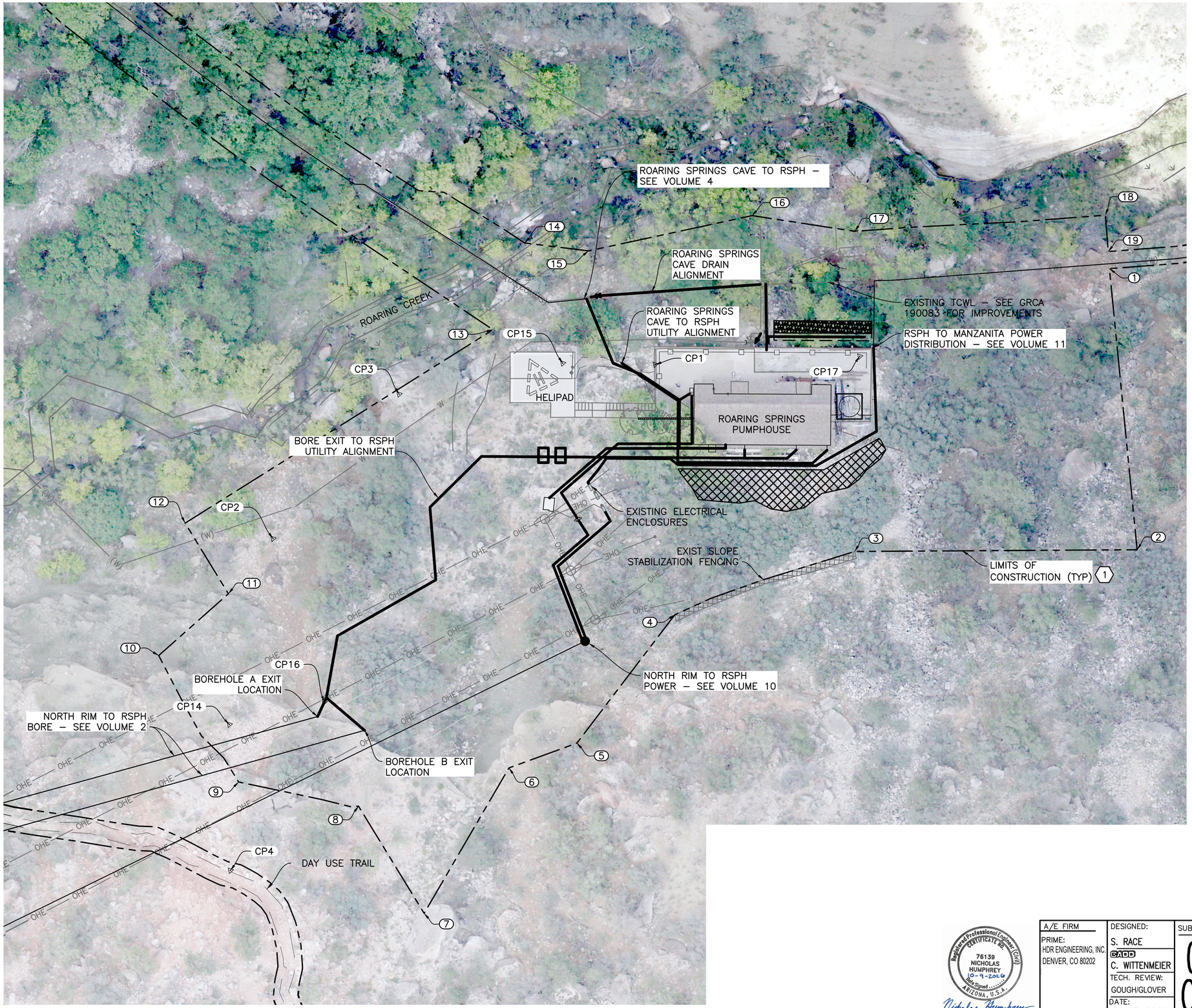

GENERAL NOTES:

A. SURVEY CONTROL AND CONTROL POINT TABLE LOCATED IN VOLUME 0. SEE SHEET 002C105.

KEYNOTES: (X)

1. AVAILBLE CONSTRUCTION LIMITS ARE SHOWN IN THE CONSTRUCTION LIMITS TABLE.

CONSTRUCTION LIMITS TABLE		
POINT NO.	NORTHING	EASTING
1	13142648.87	1335558.01
2	13142555.64	1335490.03
3	13142631.92	1335403.47
4	13142661.57	1335331.40
5	13142649.08	1335266.64
6	13142660.00	1335239.17
7	13142638.74	1335173.77
8	13142689.27	1335182.90
9	13142729.25	1335152.93
10	13142789.52	1335162.95
11	13142789.39	1335201.00
12	13142822.69	1335207.02
13	13142797.91	1335353.46
14	13142815.74	1335386.71
15	13142795.96	1335404.03
16	13142762.21	1335463.50
17	13142729.48	1335490.56
18	13142666.26	1335571.34
19	13142654.41	1335562.29

20 0 20 40
SCALE OF FEET
1" = 20'

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 10-9-2026 Date Signed ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: S. RACE CADD C. WITTENMEIER TECH. REVIEW: GOUGH/GLOVER DATE: JUNE 2026	SUB SHEET NO. 050 C101	TITLE OF SHEET GENERAL SITE PLAN NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 2 OF 113

10/10/2026 Green, Jake C:\Users\jagreen\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2022\Project Files\CAD_HDR\DRAWINGS\CIVIL\050-C101.dwg

GENERAL

G1. SCOPE
THE NOTES ON THIS SHEET AND THE STANDARD STRUCTURAL DETAILS ARE GENERAL AND APPLY TO THE ENTIRE PROJECT WHETHER SPECIFICALLY CALLED OUT OR NOT, EXCEPT WHERE THERE ARE SPECIFIC INDICATIONS TO THE CONTRARY ON STRUCTURAL SHEETS. IF THERE ARE QUESTIONS, THEY SHALL BE SUBMITTED TO THE CONTRACTING OFFICER AND ANSWERED IN WRITING PRIOR TO CONSTRUCTION.

G2. APPLICABLE SPECIFICATIONS AND CODES
A. INTERNATIONAL BUILDING CODE, IBC 2024, WITH APPLICABLE EDITIONS OF THE CODE REFERENCED STANDARDS.
B. AUTHORITY HAVING JURISDICTION: NATIONAL PARK SERVICE (WIND, SNOW, AND SEISMIC DESIGN CRITERIA IN ACCORDANCE WITH THE MORE CONSERVATIVE OF COCONINO COUNTY, AZ AND NATIONAL PARK SERVICE.)
C. CODE REQUIREMENTS FOR ENVIRONMENTAL ENGINEERING CONCRETE STRUCTURES, ACI 350-20.
D. BUILDING CODE REQUIREMENTS FOR REINFORCED CONCRETE, ACI 318-19.
E. MINIMUM DESIGN LOADS AND ASSOCIATED CRITERIA FOR BUILDINGS AND OTHER STRUCTURES, ASCE 7-22.
F. AMERICAN INSTITUTE OF STEEL CONSTRUCTION STEEL CONSTRUCTION MANUAL, 15TH ED.
G. AMERICAN WELDING SOCIETY, AWS D1.1.
H. ASTM MATERIAL STANDARDS AS NOTED.

G3. DESIGN CRITERIA
1. APPLIES TO ALL STRUCTURES (UNO)
A. DEAD LOAD:
1. ACTUAL TRIBUTARY STRUCTURE WEIGHT
2. SUPERIMPOSED DEAD LOAD: 20 PSF AT ROOF
B. LIVE LOAD:
1. ELEVATED FLOORS: 150 PSF
2. WALKWAYS, STAIRS, GRATING: 100 PSF
3. SLAB ON GRADE: 250 PSF
4. ROOF: 20 PSF (NOT REDUCIBLE)
C. WIND:
1. ULTIMATE WIND SPEED (V_{ult}): 115 MPH
2. TORNADO SPEED (V_T): NOT APPLICABLE PER ASCE 7 FIGURES 32.1-1 AND 32.1-2
3. NOMINAL WIND SPEED (V_{ASD}): 89 MPH
4. RISK CATEGORY: IV
5. EFFECTIVE AREA (A_e): 1,600 ft^2
6. EXPOSURE: C
7. INTERNAL PRESSURE COEFFICIENT (GC_{pi}): ±0.18
D. SEISMIC:
1. PUMP HOUSE FACILITY:
a. RISK CATEGORY IV
b. IMPORTANCE FACTOR (I_e): 1.50
c. SPECTRAL RESPONSE ACCELERATION, S_S = 0.42
d. SPECTRAL RESPONSE ACCELERATION, S_1 = 0.11
e. SITE CLASS: C
f. SEISMIC DESIGN CATEGORY: C
g. SPECTRAL RESPONSE COEFFICIENT, S_{DS} = 0.31
h. SPECTRAL RESPONSE COEFFICIENT, S_{D1} = 0.11
E. SNOW LOAD:
NOTE: DUE TO THE NATURE OF THE RSPH REHABILITATION, SNOW LOADS WERE TAKEN BASED ON COCONINO COUNTY LOCAL REQUIREMENTS IN LIEU OF ASCE 7-22 SNOW LOADS. THIS DECISION WAS MADE TO REFLECT THE FACT THAT THE EXISTING STRUCTURE IS NOT BEING MODIFIED TO A SIGNIFICANT ENOUGH DEGREE TO WARRANT A STRUCTURAL REDESIGN TO THE 2024 EDITION OF THE CODE. THESE MODIFICATIONS DO NOT CONTRIBUTE TO A ROOF DEAD LOAD INCREASE OF >5%.
1. GROUND SNOW LOAD (p_g) = 30 PSF
2. FLAT ROOF SNOW LOAD (p_f) = 30 PSF
3. EXPOSURE FACTOR (C_e) = 0.9
4. RISK CATEGORY = III
5. SLOPED ROOF FACTOR (C_s) = 0.82
6. THERMAL FACTOR (C_t) = 1.1
7. DRIFT SURCHARGE (p_d) = 24 PSF
8. WIDTH OF DRIFT (w) = 5'
9. WINTER WIND PARAMETER (W_2) = 0.35
10. FROST DEPTH = 24 INCHES
F. FLOOD CRITERIA: NOT APPLICABLE

G4. THE FOLLOWING NON-CONTRACTUAL GEOTECHNICAL REPORT WAS DEVELOPED FOR THIS PROJECT AND IS THE BASIS OF THIS STRUCTURAL DESIGN:
GEOTECHNICAL FIRM NAME: LITHOS ENGINEERING
ADDRESS: 2750 S. WADSWOTH BLVD., SUITE D-200 DENVER, CO 80227
REPORT NUMBER: 21186
REPORT DATE: JANUARY 30, 2024
ALLOWABLE SOIL BEARING = 2,000 PSF

G5. SAFETY
SAFETY AND STRUCTURE STABILITY DURING CONSTRUCTION ARE THE SOLE RESPONSIBILITY OF THE CONTRACTOR. STRUCTURES HAVE BEEN DESIGNED TO RESIST THE DESIGN LIVE LOADS ONLY AS A COMPLETED STRUCTURE.

G6. OPENINGS
OPENINGS FOR PIPES, DUCTS, CONDUITS, ETC. ARE NOT ALL SHOWN ON THE STRUCTURAL DRAWINGS. COORDINATE AND PROVIDE OPENINGS AS REQUIRED TO ACCOMMODATE ALL WORK SHOWN OR SPECIFIED IN THE CONTRACT DOCUMENTS AND OTHERWISE REQUIRED FOR THE FURNISHING OF A FUNCTIONALLY COMPLETE PROJECT. REINFORCE AROUND OPENINGS PER STANDARD STRUCTURAL DETAILS UNLESS OTHERWISE SHOWN.

G7. SPECIAL INSPECTIONS
SPECIAL INSPECTIONS ARE REQUIRED IN ACCORDANCE WITH CHAPTER 1 AND CHAPTER 17 OF THE IBC. PAYMENT FOR THESE INSPECTIONS IS NOT THE RESPONSIBILITY OF THE CONTRACTOR IN ACCORDANCE WITH THE REQUIREMENTS OF SPECIFICATION SECTION 01 40 00. THE CONTRACTOR SHALL PROVIDE FOR FULL ACCESS TO THE WORK BY THE SPECIAL INSPECTOR AND SHALL PROVIDE FOR THESE INSPECTIONS IN THE OVERALL CONSTRUCTION SCHEDULE. THE SPECIAL INSPECTION PROGRAM SHALL BE IN ACCORDANCE WITH SPECIFICATION SECTION 01 40 00 AND "STATMENT OF STRUCTURAL TESTS AND SPECIAL INSPECTIONS" DOCUMENTS INCLUDED AS ATTACHMENTS TO SPECIFICATION SECTION 01 40 00.

GENERAL (CONT.)

G8. STANDARD DETAILS
THE STANDARD DETAILS DEPICT TYPICAL DETAILING TO BE USED ON THIS PROJECT. IF CONDITIONS ARE NOT EXPLICITLY SHOWN ON THE DRAWINGS THEY SHALL BE MADE SIMILAR TO THE STANDARD DETAILS. OBTAIN APPROVAL OF CONTRACTING OFFICER IN WRITING FOR SIMILAR CONDITIONS PRIOR TO CONSTRUCTION.

G9. THE CONTACTOR SHALL FIELD VERIFY ALL DIMENSIONS AND ELEVATIONS OF EXISTING CONSTRUCTION AS REQUIRED TO COORDINATE NEW CONSTRUCTION. SUBMIT REQUIRED CHANGES FOR APPROVAL.

G10. CONTRACTOR TO SUBMIT FOR REVIEW ALL EQUIPMENT SIZES, OPERATING WEIGHTS, VIBRATION FORCES, SUPPORT LOCATIONS, ALONG WITH ANY FLOOR OPENINGS, NOTCHES, AND RECESSES REQUIRED BY SUCH EQUIPMENT. CONCRETE SUPPORT PADS AND/OR FRAMING REQUIRED TO SUPPORT SAID EQUIPMENT SHALL NOT BE FABRICATED AND PLACED UNTIL THE CONCRETE SUPPORT PADS AND/OR FRAMING IS APPROVED TO SUPPORT THE EQUIPMENT.

CONCRETE

C1. DESIGN STRENGTHS:
F'c = 4,500 PSI (UNO)
F'c = 5,000 PSI (ALL EXTERIOR CONCRETE)
Fy = 60,000 PSI REINFORCEMENT BARS - ASTM A615, GR 60

C2. CONCRETE COVER
UNLESS OTHERWISE NOTED, PROVIDE CONCRETE COVER FOR REINFORCING AS FOLLOWS:
CONCRETE DEPOSITED AGAINST EARTH: 3"
ALL OTHER: 2"
SEE DRAWINGS FOR EXCEPTIONS

C3. SEE SPECIFICATIONS FOR REINFORCING PLACEMENT REQUIREMENTS.

C4. REFER TO OTHER DISCIPLINE DRAWINGS PRIOR TO CONSTRUCTION FOR EMBEDDED ITEMS AND PENETRATIONS NOT SHOWN ON STRUCTURAL DRAWINGS AS REQUIRED TO ACCOMMODATE ALL WORK SHOWN OR SPECIFIED IN THE CONTRACT DOCUMENTS AND OTHERWISE REQUIRED FOR THE FURNISHING OF A FUNCTIONALLY COMPLETE PROJECT. REINFORCE AROUND OPENINGS PER STANDARD STRUCTURAL DETAILS UNLESS OTHERWISE SHOWN.

C5. PROVIDE 3/4" CHAMFERS AT ALL EXPOSED EDGES. NOT ALL CHAMFERS MAY BE SHOWN ON DRAWINGS.

C6. FIELD ADJUST REINFORCING AT OPENINGS AND EMBEDDED ITEMS AS INDICATED.

C7. ANCHOR BOLTS NOT SPECIFIED BY ENGINEER SHALL BE DESIGNED AND CERTIFIED BY A REGISTERED PROFESSIONAL ENGINEER LICENSED IN THE STATE OF ARIZONA, RETAINED BY THE CONTRACTOR, IN ACCORDANCE WITH APPLICABLE PROJECT AND CODE REQUIREMENTS. SUBMIT AS A SHOP DRAWING FOR REVIEW AND APPROVAL BY THE CONTRACTING OFFICER. COORDINATE LOCATION, SIZE AND EMBEDMENT PRIOR TO CASTING CONCRETE.

C8. CONTINUOUS WATERSTOP SHALL BE INSTALLED IN JOINTS SUBJECT TO STATIC WATER PRESSURE.

C9. ABSOLUTELY NO WELDING OF REINFORCING BARS OR TORCHING TO BEND REINFORCING BARS SHALL BE ALLOWED WITHOUT SPECIFIC APPROVAL FROM THE CONTRACTING OFFICER.

C10. CONTRACTOR SHALL SUBMIT A CONCRETE PLACEMENT PLAN{PER SPECIFICATIONS IDENTIFYING JOINT TYPES, JOINT LOCATIONS AND CONCRETE PLACEMENT SEQUENCE.

C11. ALL CAST IN PLACE AND POST-INSTALLED ANCHORS INDICATED IN THE STRUCTURAL DOCUMENTS SHALL COMPLY WITH CHAPTER 17 OF ACI 318, APPENDIX E OF ACI 350 AND CHAPTER 19 OF THE IBC. ALL EXPANSION AND ADHESIVE ANCHORS SHALL HAVE THE ICC REPORT SHOWING EQUIVALENT LOAD CAPACITY. SUBMIT AND INSTALL PER THE ICC EVALUATION REPORT.

STEEL

S1. DESIGN STRENGTHS, UNO:
WIDE FLANGE AND TEES - ASTM A992: Fy=50 KSI
PIPES - ASTM A53, GRADE B: Fy=35 KSI
STAINLESS STEEL, TYPE 304 OR 316 AND TYPE 304L AND 316L WHERE WELDING IS REQUIRED: Fy=33 KSI
HSS SECTIONS, ASTM A500 GRADE C: Fy=50 KSI
ALL OTHER PLATES AND SHAPES, ASTM A36: Fy=36 KSI
ANCHOR RODS, ASTM F1554 GRADE 36: Fy=36 KSI

S2. DIMENSIONS:
TO CENTERLINES OF COLUMNS AND BEAMS, TOP SURFACES OF BEAMS AND TUBES AND BACKS OF CHANNELS AND ANGLES UNO.

S3. ELEVATIONS:
TOP OF STEEL REFERS TO TOP SURFACE OF MEMBER OR FLANGE UNO.

S4. WHEN FILLET WELD SIZE IS NOT INDICATED, PROVIDE MAXIMUM WELD SIZE BASED ON MATERIAL THICKNESS IN ACCORDANCE WITH AISC SPECIFICATIONS.

S5. ALL BOLTED STRUCTURAL CONNECTIONS ARE BEARING TYPE CONNECTIONS UNLESS OTHERWISE SPECIFIED TO BE SLIP-CRITICAL. PROVIDE LOAD INDICATING WASHERS AT SLIP-CRITICAL CONNECTIONS.

S6. CONFORM TO AISC 360, STEEL CONSTRUCTION MANUAL.

ALUMINUM

A1. STRUCTURAL ALUMINUM YIELD STRENGTHS
STRUCTURAL ALUMINUM: Fy=35 KSI
STRUCTURAL ALUMINUM IS ALLOY 6061-T6 UNO

A2. DIMENSIONS:
TO CENTERLINES OF COLUMNS AND BEAMS, TOP SURFACES OF BEAMS AND TUBES AND BACKS OF CHANNELS AND ANGLES UNO.

A3. ELEVATIONS:
TOP OF ALUMINUM REFERS TO TOP SURFACE OR FLANGE OF MEMBER UNO.

A4. WHEN FILLET WELD SIZE IS NOT INDICATED, PROVIDE MAXIMUM WELD SIZE FOR THE MATERIAL THICKNESS IN ACCORDANCE WITH THE LATEST EDITION OF THE "ALUMINUM DESIGN MANUAL" BY THE ALUMINUM ASSOCIATION.

A5. ALUMINUM IN CONTACT WITH DISSIMILAR MATERIALS OR CONCRETE:
CONTACT SURFACES SHALL BE PROVIDED WITH GALVANIC SEPERATION PER SPECIFICATIONS.

DELEGATED DESIGN SUBMITTALS:

D1. DEFINITION:

DELEGATED DESIGN SUBMITTALS ARE ITEMS FOR WHICH THE FINAL DESIGN OF A SPECIFIC ELEMENT OF THE PERMANENT CONSTRUCTION WORK EXPRESSLY DELEGATED TO A PROPERLY LICENSED AND REGISTERED DESIGN PROFESSIONAL RETAINED OR EMPLOYED BY THE CONTRACTOR, SUBCONTRACTOR, SUPPLIER OF ANY TIER. "DELEGATED DESIGN" APPLIES ONLY TO PERMANENT WORK THAT WILL BE PART OF THE COMPLETED PROJECT AS A FUNCTIONING WHOLE.

D2. DELEGATED DESIGN LIST:

THE FOLLOWING IS A LIST OF DELEGATED DESIGN SUBMITTALS THAT ARE EXPECTED TO CONTAIN STRUCTURAL CALCULATIONS OF SAFETY-RELATED SYSTEM INFORMATION FOR REVIEW. THESE SUBMITTALS APPLY TO DELEGATED DESIGNED SYSTEMS PRIOR TO INSTALLATION OF THE INDICATED STRUCTURAL ELEMENT, EQUIPMENT, DISTRIBUTION SYSTEM, COMPONENT, OR ITS ANCHORAGE. THE CONTRACTOR SHALL SUBMIT THE REQUIRED ENGINEER CERTIFICATION, SUPPORTING DATA, AND DRAWINGS FOR REVIEW AND ACCEPTANCE BY THE CONTRACTING OFFICER.

D3. DELEGATED DESIGN SUBMITTALS INCLUDE:
1. GUARDRAIL
2. GRATING AND ASSOCIATED SUPPORTS AND CONNECTIONS
3. PRECAST CONCRETE ELEMENTS, INCLUDING MODULAR BLOCK WALL

REGISTERED PROFESSIONAL ENGINEER — CERTIFICATE NO. 85509 — JOHN CORONADO — This item is Signed and Sealed electronically — ARIZONA U.S.A.

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: Z. NANNEN DRAWN: R. NELSON TECH. REVIEW: J. CORONADO DATE: MAY 2026	SUB SHEET NO. 050 S001	TITLE OF SHEET GENERAL GENERAL STRUCTURAL NOTES NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 3 OF 113

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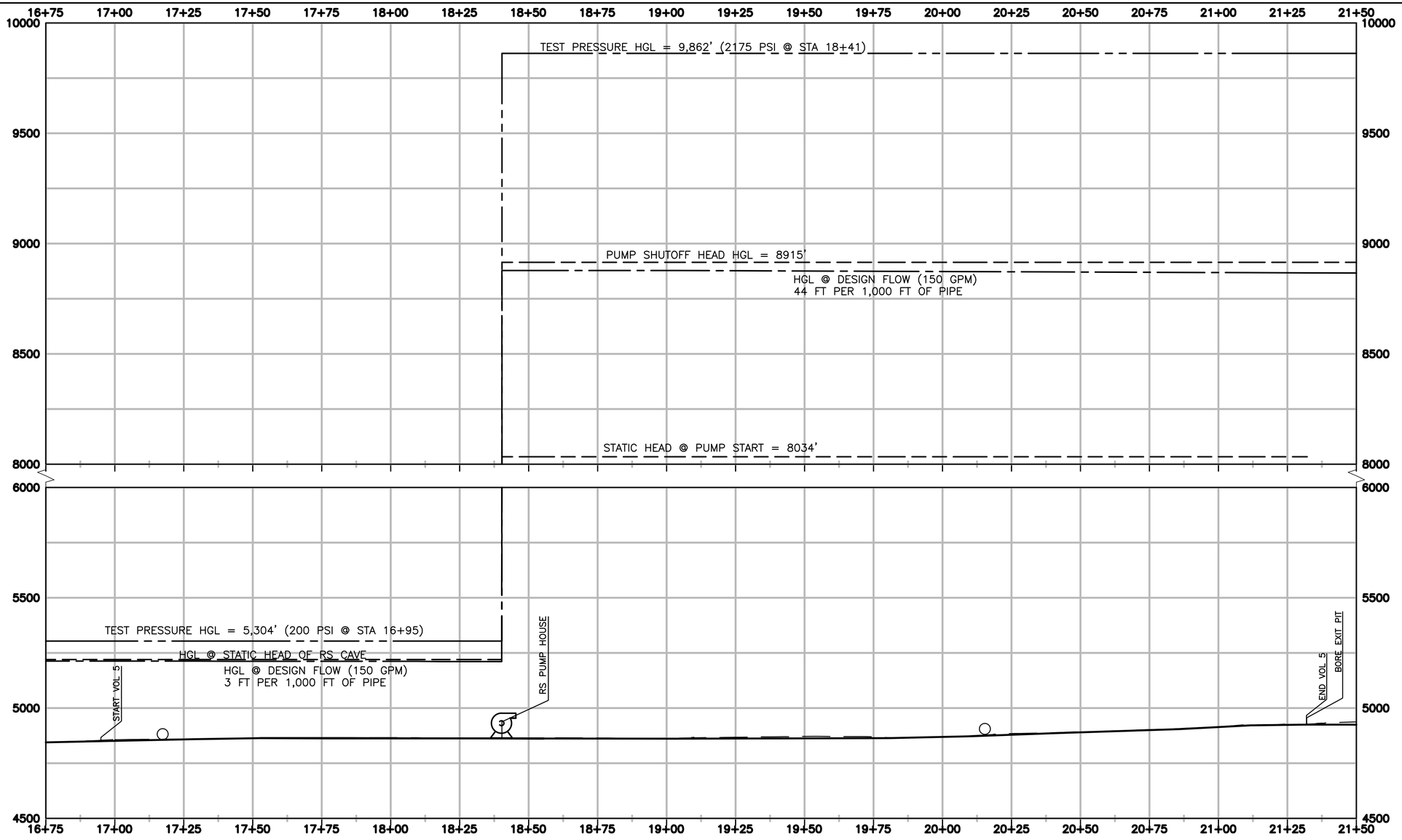


ROARING SPRINGS PUMPHOUSE TO BORE EXIT

GENERAL NOTES:

A. CONTRACTOR TO SUBMIT TESTING PLAN TO COR PRIOR TO TESTING OF THE STATIC LINE.

B. FOR FULL PIPELINE LENGTH HYDRAULIC PROFILES SEE VOLUME 0 SHEET 005/D101.

LEGEND:

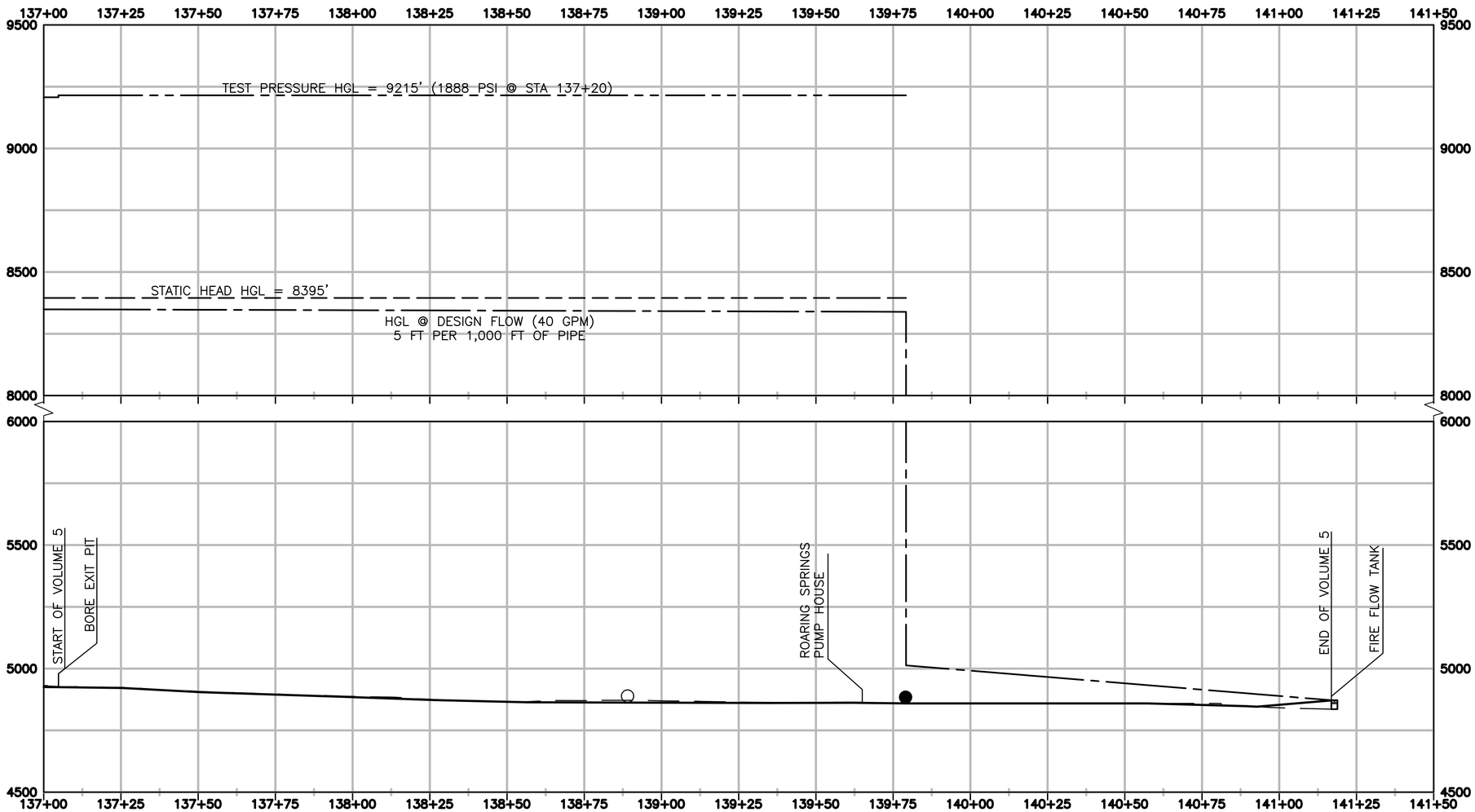
- PUMP
- ISOLATION VALVE
- EXISTING GROUND AT SURFACE
- WATERLINE HGL
- PIPE CENTERLINE
- PUMP SHUTOFF HEAD
- TRANSIENT HGL
- TEST PRESSURE
- STATIC HEAD

20 0 20 40 HORIZ.
250 0 250 500 VERT.
SCALE OF FEET
H: 1"=20' V: 1"=250'

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 9-1-2026 Date Signed ARIZONA, U.S.A.
Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: N. HUMPHREY CADD C. WITTENMEIER TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 050 D101	TITLE OF SHEET GENERAL RAW WATER HYDRAULIC PROFILE NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 4 OF 113

4/17/2026 Willyard, Lisa C:\Users\LWILLYARD\DC\ACCDocs\HDR\10350545_NPS_GRCA_N_Rim_Utility_2026\Project Files\CAD_HDR\DRAWINGS\PROCESS\050-D101.dwg



BORE EXIT TO ROARING SPRINGS PUMPHOUSE

GENERAL NOTES:

A. CONTRACTOR TO SUBMIT TESTING PLAN TO COR PRIOR TO TESTING OF THE STATIC LINE.

B. FOR FULL PIPELINE LENGTH HYDRAULIC PROFILES SEE VOLUME 0 SHEET 005/D102.

LEGEND:

- ISOLATION VALVE
- PRESSURE REDUCING VALVE
- FIRE FLOW TANK
- EXISTING GROUND AT SURFACE
- PIPE CENTERLINE
- WATERLINE HGL
- TEST PRESSURE
- STATIC HEAD

20 0 20 40 HORIZ.
250 0 250 500 VERT.
SCALE OF FEET
H: 1"=20' V: 1"=250'

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A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: N. HUMPHREY DRAWN: C. WITTENMEIER TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 050 D102	TITLE OF SHEET GENERAL FINISHED WATER HYDRAULIC PROFILE NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 5 OF 113

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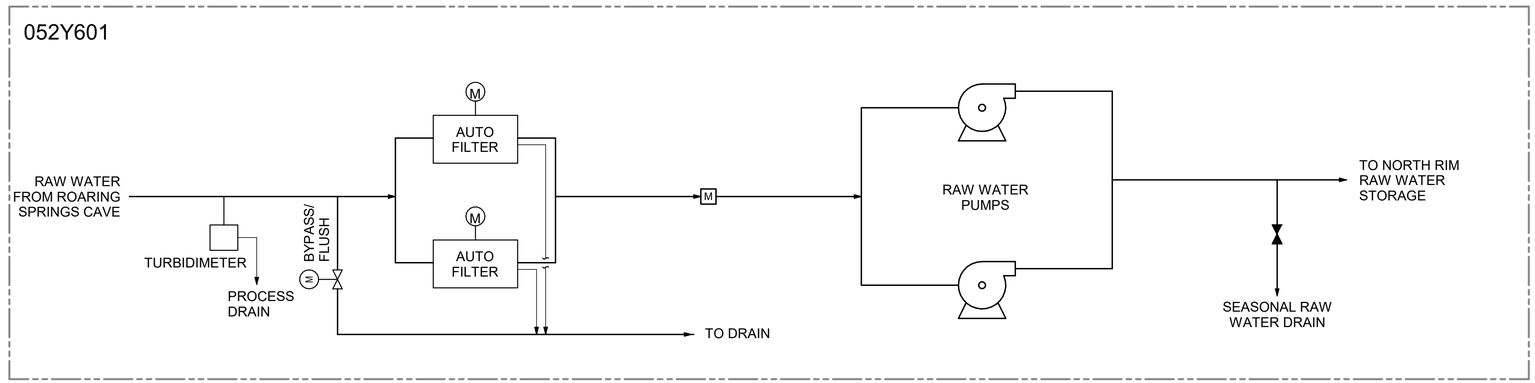
GENERAL NOTES

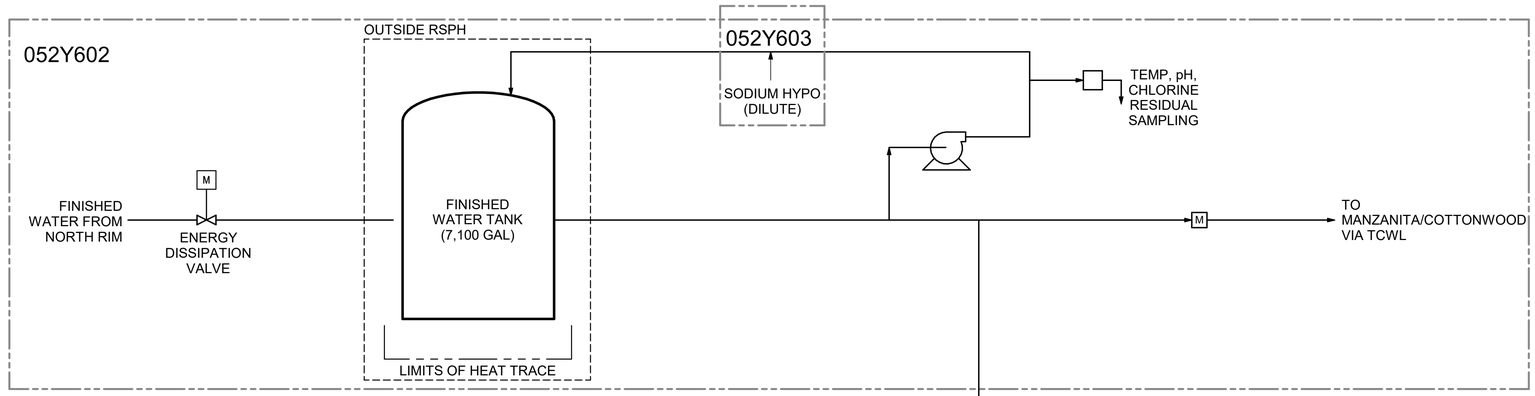
1. ALL PROCESS COMPONENTS LOCATED IN RSPH EXCEPT WHERE NOTED.

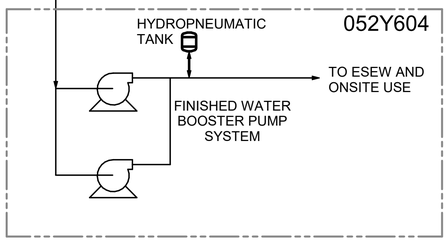
LEGEND

LIQUID PROCESS FLOW

P&ID'S







Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: D. ZINCHENKO DRAWN: N. LYNCH TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 050 D601	TITLE OF SHEET GENERAL PROCESS FLOW DIAGRAM NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 6 OF 113

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PUMP STATION FLOW RATES

RAW WATER FEED RATE	150 GPM
FINISHED WATER FEED RATE (MIN/MAX)	40 GPM
MAX DEMAND - FIRE WATER	132 GPM

RAW WATER PUMPS

NUMBER OF PUMPS	2 (1 DUTY + 1 STANDBY)
PUMP TYPE	MULTISTAGE HORIZONTAL SPLIT CASE
DRIVE TYPE	VFD
DESIGN CAPACITY (EA)	150 GPM
TDH AT DESIGN CAPACITY	3,604 FT
MOTOR TYPE	THREE-PHASE ELECTRIC OPEN DRIP ROOF
HORSEPOWER	450

AUTO FILTERS

NUMBER OF FILTERS	2
MAXIMUM RATED OPERATING PRESSURE	200 PSI
MAXIMUM FLOW PER UNIT	250GPM
MAXIMUM HEADLOSS	7 PSI
MICRON RATING	10

FINISHED WATER TANK

TYPE	FIBERGLASS REINFORCED PLASTIC
NUMBER	1
DIAMETER	11 FT
HEIGHT	11 FT
MAX OPERATING TANK VOLUME (TOTAL)	7100 GAL

HYDRO-PNEUMATIC TANKS (BLADDER STYLE)

NUMBER	1
TANK VOLUME	53 GAL
OPERATIONAL VOLUME (EFFECTIVE)	17 GAL
RATED OPERATING PRESSURE	50 PSI

RECIRCULATING PUMP

NUMBER OF PUMPS	1
PUMP TYPE	MAGNETIC DRIVE CENTRIFUGAL
DRIVE TYPE	CONSTANT SPEED
DESIGN CAPACITY (EA)	20 GPM
TDH AT DESIGN CAPACITY	20 FT
MOTOR TYPE	SINGLE PHASE, TEFC
HORSEPOWER	0.12 - 0.32 HP

BOOSTER PUMPS

NUMBER OF PUMPS	2 (1 DUTY + 1 STANDBY)
PUMP TYPE	VERTICAL INLINE
DRIVE TYPE	CONSTANT SPEED
DESIGN CAPACITY (EA)	20 GPM
TDH AT DESIGN CAPACITY	185 FT
MOTOR TYPE	SINGLE PHASE, TEFC
HORSEPOWER	1.5 HP

CHEMICAL FEED SYSTEMS

SODIUM HYPOCHLORITE

TYPE	LIQUID
STOCK SOLUTION	
ACTIVE STRENGTH	12.5% (W/W)
STORAGE VOLUME	5 OR 15 GAL
6-MONTH SUPPLY	1 GAL
DILUTE SOLUTION	
ACTIVE STRENGTH	0.5 - 1.0%
STORAGE VOLUME	60 GAL
BATCH METERING PUMPS	
APPLICATION POINT	DILUTE SOLUTION STORAGE TANK
DOSAGE (TYP)	5,000 - 10,000 MG/L (0.5-1.0%)
NUMBER OF VENDOR SUPPLIED PUMPS	2 (1 DUTY + 1 SHELF SPARE)
PUMP CAPACITY	0.001 - 7.67 GPH
CHEMICAL METERING PUMPS	
APPLICATION POINT	FW TANK RECIRCULATING LINE
TARGET CHLORINE CONCENTRATION	1.5 MG/L
MINIMUM DAILY CHLORINE DEGRADATION	0.1 MG/L
MAXIMUM DAILY CHLORINE DEGRADATION	0.5 MG/L
NUMBER OF VENDOR SUPPLIED PUMPS	2 (1 DUTY + 1 STANDBY)
PUMP CAPACITY	0.001 - 7.67 GPH

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: R. STOUGES CADD N. LYNCH TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 050 D602	TITLE OF SHEET GENERAL DESIGN CRITERIA NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 7 OF 113	

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GENERAL NOTES:

A. THE SCHEDULE APPLIES TO INTERIOR PROCESS PIPING INSIDE RSPH. SEE CIVIL SHEETS FOR BURIED PIPING REQUIREMENTS.

PIPE SERVICE	DESCRIPTION	SPECIFICATION SECTION	SIZE	MATERIAL	CLASS OR WALL THICKNESS	INTERIOR LINING	EXTERIOR COATING	PRESSURE TESTING REQUIREMENTS			COLOR CODING
DR	MISC INSTRUMENT DRAIN	40 05 31	≤2"	PVC	SCH 40 OR 80	NONE	NONE	MEDIUM: WATER	DURATION: 1.5 HRS	TEST PRESSURE: 10 PSI	NONE
HYPO	SODIUM HYPOCHLORITE	40 05 31	3/8" (OD)	PVC TUBING	≥1/16" THK	NONE	NONE	MEDIUM: WATER	DURATION: 4 HRS	TEST PRESSURE: 50 PSI	YELLOW BANDS
			≥1/2"	CPVC	SCH 80						
HYPO-DL	DILUTE HYPOCHLORITE	40 05 31	3/8" (OD)	PVC TUBING	≥1/16" THK	NONE	NONE	MEDIUM: WATER	DURATION: 4 HRS	TEST PRESSURE: 50 PSI	YELLOW AND BLUE BANDS
			≥1/2"	CPVC	SCH 80						
PD1	PROCESS DRAIN	40 05 24	≤6"	STL	SCH 40	NONE	HPIC, SPEC 09 96 00	MEDIUM: WATER	DURATION: 6 HRS	TEST PRESSURE: 180 PSI	LIGHT BROWN COATING
PD2	HIGH VELOCITY PROCESS DRAIN	40 05 24	≤4"	STL	SCH 80	NONE	HPIC, SPEC 09 96 00	MEDIUM: WATER	DURATION: 6 HRS	TEST PRESSURE: 570 PSI	DARK BROWN COATING
RW4	RAW WATER	40 05 24	4"	STL	SCH 40	NONE	HPIC, SPEC 09 96 00	MEDIUM: WATER	DURATION: 6 HRS	TEST PRESSURE: 180 PSI	LIGHT GREEN COATING
RW5	RAW WATER - HIGH PRESSURE	40 05 24	4"	STL	SCH 80	NONE	HPIC, SPEC 09 96 00	MEDIUM: WATER	DURATION: 6 HRS	TEST PRESSURE: 2,225 PSI	DARK GREEN COATING
W1	POTABLE / FINISHED WATER	40 05 24	≤2"	STL (downstream of W5)	SCH 40	NONE	HPIC, SPEC 09 96 00	MEDIUM: WATER	DURATION: 6 HRS	TEST PRESSURE: 160 PSI	LIGHT BLUE COATING
		40 05 31	≤2"	PVC	SCH 80		NONE			TEST PRESSURE: 105 PSI	LIGHT BLUE BANDS
			≥2.5"							TEST PRESSURE: 10 PSI	
		33 05 33.23	3" - 4"	HDPE	DR 9					TEST PRESSURE: 10 PSI	
W5	POTABLE / FINISHED WATER - HIGH PRESSURE	40 05 24	≤4"	STL	SCH 80	NONE	HPIC, SPEC 09 96 00	MEDIUM: WATER	DURATION: 6 HRS	TEST PRESSURE: 2,225 PSI	DARK BLUE COATING

Registered Professional Engineer (Civil) CERTIFICATE NO. 76139 NICHOLAS HUMPHREY 5-1-2026 Date Signed ARIZONA, U.S.A.

Nicholas Humphrey

A/E FIRM PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	DESIGNED: R. STOUGES DRAWN: N. LYNCH TECH. REVIEW: GOUGH/GLOVER DATE: MAY 2026	SUB SHEET NO. 050 D603	TITLE OF SHEET GENERAL PIPE SCHEDULE NORTH RIM - GRAND CANYON NATIONAL PARK	DRAWING NO. 113 182752 PMIS/PKG NO. 318719 SHEET 8 OF 113

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RSPH PLC CP

MANAGED ETHERNET SWITCH

RSPH MSW

120 VAC ROUTER (OSPF) RSPH RTR2

120 VAC ROUTER (OSPF) RSPH RTR1

CRYPTO

24VDC

FIBER PATCH PANEL

24 STRAND SINGLEMODE FIBER

FOPP52

144 STRAND SM FIBER
24 STRAND ARMORED SM FIBER
TO NR WTP

UPS

RSPH UPS

RSPH HMI

CAMERA
CAMERA
CAMERA

NETWORK TIME PROTOCOL SERVER

24VDC

3

VFD-52211
VFD-52212
MCC SPD
MCC DMP

RS CAVE CP

MEDIA CONVERTER

2

24 VDC

CAT6

RADIO

WIRELESS RADIO

TO MANZANITA POINT

FIBER PATCH PANEL

MANAGED ETHERNET SWITCH

CAMERA
CAMERA

24VDC

RS CAVE MSW

MEDIA CONVERTER

24 VDC

RSPH RIO 4

120VAC	D4-08B-1 120VAC	H4-EBC	F4-08RTD RTD 8 CH RTD	F2-08TRS-1 RELAY 8 PT DO	F4-16AD-1 4-20mA 16 CH AI	D4-FILL	D4-FILL	D4-FILL	D4-FILL	D4-FILL
CAT6		E								
BASE		0	1	2	3	4	5	6	7	

RSPH MASTER PLC

120VAC	D2-04B-1 120VAC	H2-DM1E PLC CPU	H2-ERM100	H2-ECOM	D2-08CDR 24VDC/RELAY 4 PT DI/4 PT DO
		S			
		E	E	E	
BASE			0	1	2

UNMANAGED ETHERNET SWITCH

RSPH SW

450 MHZ RADIO

24 VDC

1

WIRELESS RADIO

TO NR WTP

RSPH STANDBY PLC

120VAC	D2-04B-1 120VAC	H2-DM1E PLC CPU	H2-ERM100	H2-ECOM	D2-08CDR 24VDC/RELAY 4 PT DI/4 PT DO
		S			
		E	E	E	
BASE			0	1	2

LCP-53117

120 VAC

UPS 2200VA

LCP UPS

MANAGED ETHERNET SWITCH

CAT6

CAT6

TO EDS-53117

6 STRAND SINGLEMODE FIBER

CAT6

LIC

24VDC

LCP MSW

RSPH RIO 1

120VAC	D2-09B-1 120VAC	H2-EBC100							
		E							
BASE		0	1	2	3	4	5	6	7

RSPH RIO 2

120VAC	D2-09B-1 120VAC	H2-EBC100							
		E							
BASE		0	1	2	3	4	5	6	7

RSPH RIO 3

120VAC	D2-09B-1 120VAC	H2-EBC100							
		E							
BASE		0	1	2	3	4	5	6	7

GENERAL NOTES:

A. DRAWING IS DIAGRAMATIC.

B. EXISTING ETHERNET NETWORK IS ASSUMED TO BE STABLE, SECURE, & FULLY OPERATIONAL.

C. REFERENCE SPECS FOR DETAILS.

D. IP ADDRESSES WILL BE ISSUED AFTER AWARD OF CONSTRUCTION.

E. SEPARATE VLANS SHALL BE UTILIZED FOR THE SCADA NETWORK, VFD'S, AND SECURITY CAMERAS.

F. REFERENCE SECTION 40 61 93 FOR I/O MODULES ON EACH PLC/RIO RACK.

G. SALVAGE EXISTING NETWORK DEVICES (INCLUDING BUT NOT LIMITED TO RADIOS, SWITCHES, PLC RACK AND CARDS) AT RSPH AND RETURN TO CONTRACTION OFFICER'S REPRESENTATIVE.

KEYNOTES (X)

1 EXISTING RADIO TO BE RELOCATED TO NEW RSPH CONTROL PANEL.

2 POE INJECTING MEDIA CONVERTER.

3 PROVIDE WITH GPS CABLE LONG ENOUGH TO REACH OUTSIDE WALL OF BUILDING.

DEVICE NAME

S - SERIAL - RS-232

& E - ETHERNET 10/100 RJ-45

FO - FIBER OPTIC PORT

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Registered Professional Engineer (Electrical) CERTIFICATE NO. 85305 JOSEPHINE A BRAINARD Date Signed 04/27/26 ARIZONA U.S.A.

A/E FIRM	DESIGNED:	SUB SHEET NO.	TITLE OF SHEET	DRAWING NO.
PRIME: HDR ENGINEERING, INC. DENVER, CO 80202	J. BRAINARD	050 Y601	GENERAL SCADA NETWORK BLOCK DIAGRAM	113 182752
	S. BILD			PMIS/PKG NO. 318719
	TECH. REVIEW: K. CHANDRA			SHEET 9 OF 113
	DATE: MAY 2026		NORTH RIM - GRAND CANYON NATIONAL PARK	